Three models, one recognition task

TRAFFIC SIGN RECOGNITION | MODEL COMPARISON & EXPERIMENT RESULTS

This report compares the implemented Baseline CNN, CNN + Adaptive Batch Normalization (AdaBN), and CNN + Ant Colony Optimization (ACO). It uses the current local source code, nine saved robustness-result CSVs, and nine original clean-test confusion-matrix PDFs.

Main result

CNN + AdaBN has the highest saved mean corruption accuracy: 89.02%, followed by CNN + ACO at 73.56% and Baseline CNN at 70.41%. These averages weight the three datasets equally.

Datasets are evaluated independently

Dataset	Region	Output classes
GTSRB	Germany	43
BTSD	Belgium	62
CTSD	China	58

Each model is trained and evaluated separately for each dataset. Class IDs belong to their own dataset; the datasets are not merged into a common label space.

What is included

Pages 2-3 explain architecture and model differences. Page 4 compares numerical metrics. Page 5 explains interpretation and limitations. Pages 6-14 reproduce one clean-test confusion matrix for every model-dataset pair.

Important distinction

Numerical tables summarize corrupted-image testing. The embedded matrices show clean-image testing. They describe different conditions and must not be treated as the same evaluation. No clean-test numerical accuracy is inferred from image colors.

Shared CNN architecture and differences

All models take RGB images resized to 49 x 49 pixels and produce one raw score (logit) per class. Pixel values are scaled to [0, 1]. The current activation is ReLU, not GELU.

Stage	Operation	Output per image
Input	RGB image	3 x 49 x 49
Block 1	3 x 3 convolution: 3 to 32; ReLU; 2 x 2 max pooling	32 x 24 x 24
Block 2	3 x 3 convolution: 32 to 64; ReLU; Dropout2d (0.2); max pooling	64 x 12 x 12
Block 3	3 x 3 convolution: 64 to 128; ReLU; max pooling	128 x 6 x 6
Classifier	Flatten; Linear 4608 to 512; ReLU; Dropout (0.5); Linear 512 to K	K logits

All convolutions use stride 1 and padding 1. Pooling halves spatial size with rounding down. K is 43, 62, or 58 depending on the dataset. Prediction is the class with the largest logit; softmax is not needed to select that class.

What changes between the models?

Property	Baseline	AdaBN	ACO
BatchNorm layers	None	After each convolution and first linear layer	None
Feature gate	None	None	128-channel pheromone gate after third pooling
Extra procedure	None	Update BN statistics on unlabeled target images	Offline ant-colony search during training
Test-time changes	None	Running mean and variance only	None; gate is fixed

ReLU keeps positive activations and sets negative ones to zero. Pooling reduces spatial resolution. Dropout randomly suppresses activations during training and is disabled during inference. These mechanisms are shared rather than unique contributions.

How each model learns and predicts

1. Baseline CNN

Convolutions learn image features and the dense classifier maps them to sign classes. Training updates the network weights using labeled images. Testing uses the saved weights without normalization adaptation or feature-search steps. This is the reference model.

2. CNN + AdaBN

BatchNorm normalizes activations using a mean and variance, then applies learned scale and offset parameters. AdaBN recalibrates the running statistics using unlabeled target images; it does not retrain the convolution or classifier weights.

For each corruption condition, the evaluator restores the source checkpoint, resets BN statistics, and passes the whole corrupted test set through the model without gradients. BN layers collect statistics while dropout stays disabled. The model is then evaluated on that condition. Clean testing occurs without this adaptation step.

Example: brightness changes may shift feature statistics. AdaBN attempts to account for the new distribution rather than using only the original training statistics.

3. CNN + ACO

The network first trains for one warm-up epoch. Before epoch 2, ACO evaluates candidate binary masks of the final 128 feature channels. The configured defaults are 15 ants, 5 search iterations, and pheromone evaporation of 0.15.

Search fitness is classification accuracy on one sampled training batch. The heuristic is normalized channel-activation variance. Pheromones are converted into a continuous gate of approximately 0.5 to 1.5, then the network continues training with the fixed gate. The gate is applied before the dense classifier and is saved with the checkpoint.

This is channel reweighting, not actual channel removal or a smaller CNN. ACO is not the optimizer used for the convolution weights, and the existing search does not explicitly optimize corruption robustness.

Shared training configuration

Source-code defaults: Adam optimizer, learning rate 0.001, cross-entropy loss, batch size 64, and 5 epochs. Defaults are not a complete audit trail of every saved run. All models use Adam for trainable weights; ACO adds its separate channel-search step.

Evaluation metrics and measured results

Every model-dataset CSV contains 14 corruption types x 5 severity levels = 70 conditions. The values below are arithmetic means of the saved per-condition metrics, not clean-test scores and not metrics pooled over one combined prediction list.

Model	Dataset	Acc. %	P micro %	R micro %	F1 macro
Baseline CNN	GTSRB	70.65	70.65	70.65	0.6702
Baseline CNN	BTSD	73.75	73.75	73.75	0.5273
Baseline CNN	CTSD	66.83	66.83	66.83	0.5045
CNN + AdaBN	GTSRB	78.79	78.79	78.79	0.7661
CNN + AdaBN	BTSD	92.30	92.30	92.30	0.8323
CNN + AdaBN	CTSD	95.98	95.98	95.98	0.9488
CNN + ACO	GTSRB	71.70	71.70	71.70	0.6891
CNN + ACO	BTSD	74.29	74.29	74.29	0.5220
CNN + ACO	CTSD	74.71	74.71	74.71	0.6301

Equal-weight average across the three datasets

Model	Acc. %	P micro %	R micro %	F1 macro
Baseline CNN	70.41	70.41	70.41	0.5673
CNN + AdaBN	89.02	89.02	89.02	0.8491
CNN + ACO	73.56	73.56	73.56	0.6137

Definitions and availability

Accuracy = correct predictions / all predictions. Precision = true positives / (true positives + false positives). Recall = true positives / (true positives + false negatives). F1 balances precision and recall through their harmonic mean.

Micro averaging aggregates class counts before calculating the metric. In this single-label, full-class classification setting, micro-precision and micro-recall both equal accuracy. Their entries are mathematically derived from saved accuracy, not independently recorded precision/recall measurements.

Macro-F1 averages per-class F1 scores and is saved on the 0-1 scale. Macro-precision and macro-recall were not saved and cannot be recovered from accuracy and macro-F1. They require another inference pass, not retraining. The evaluator uses sklearn's default class set for macro-F1, rather than explicitly fixing all dataset class IDs.

Interpretation, matrix guide and evidence

What the saved results support

AdaBN leads on mean corruption accuracy for each of the three datasets. Its equal-dataset average exceeds baseline by 18.61 percentage points and ACO by 15.46 points. ACO improves mean accuracy over baseline on all three datasets, but its BTSD macro-F1 is lower. Higher accuracy does not necessarily mean better performance across classes.

How to read the confusion matrices

Rows are true classes and columns are predicted classes. Diagonal cells represent correct classifications. Off-diagonal cells represent mistakes: a cell at row 3, column 5 means true class 3 was predicted as class 5. The following nine figures reproduce the saved clean-test matrices; they are not generated or estimated illustrations.

The matrices display raw counts through color, without cell-number annotations or a colorbar. Plot colors are independently scaled, so do not compare darkness across different matrices as a shared numerical scale. Class numbers are dataset-specific.

Limits of this comparison

The baseline has no BN, whereas the AdaBN architecture includes BN and adaptation. An additional evaluation of the same BN-equipped checkpoint with adaptation disabled is needed to isolate the adaptation benefit. These are saved single-run results, not multi-seed estimates. Historical files do not prove that all runs used identical random corruptions; shared caching was added during this experiment period.

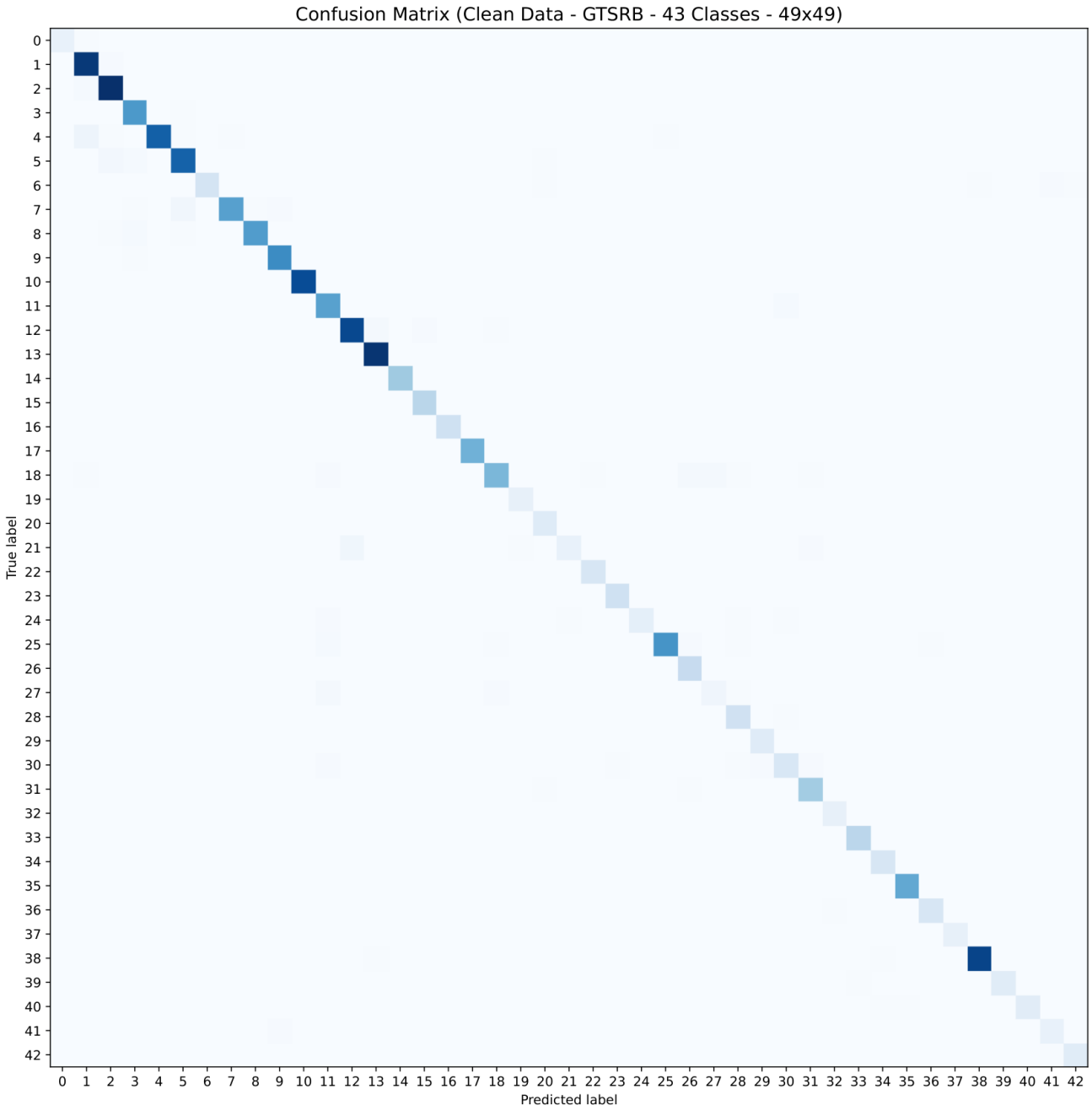
Result provenance

Root: D:/python/ML-Project. Numeric sources: output/<model>/<dataset>/robustness_results_<dataset>.csv. SHA-256 prefixes below identify the exact CSV snapshots.

Model	Dataset	CSV SHA-256 prefix
Baseline CNN	GTSRB	d6b0b500d422
Baseline CNN	BTSD	9db1189e7729
Baseline CNN	CTSD	fcf60b8d16d2
CNN + AdaBN	GTSRB	201d404edf94
CNN + AdaBN	BTSD	2dcb4b7f250b
CNN + AdaBN	CTSD	8cc2b3379519
CNN + ACO	GTSRB	e3bc62e0da86
CNN + ACO	BTSD	c9a8aa58a88d
CNN + ACO	CTSD	2c7141bb4d1e

Baseline CNN | GTSRB confusion matrix

CLEAN TEST DATA | 43 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

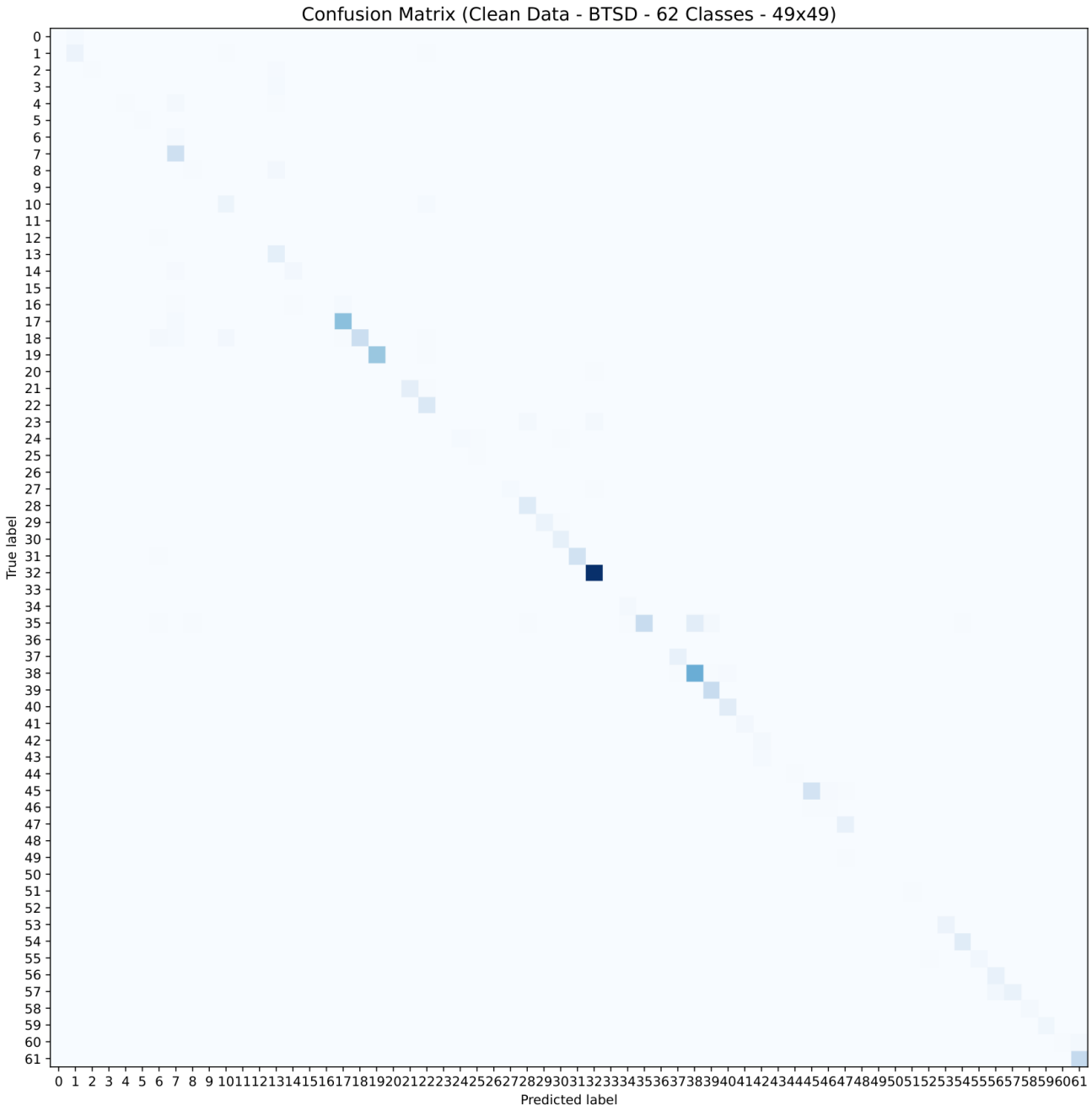
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/baseline/gtsrb/test_without_noise_gtsrb.pdf | SHA-256 prefix: ef6701890b4a

Additional corruption matrices are saved separately in output/baseline/gtsrb/noise/ (14 severity-5 PDFs).

Baseline CNN | BTSD confusion matrix

CLEAN TEST DATA | 62 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

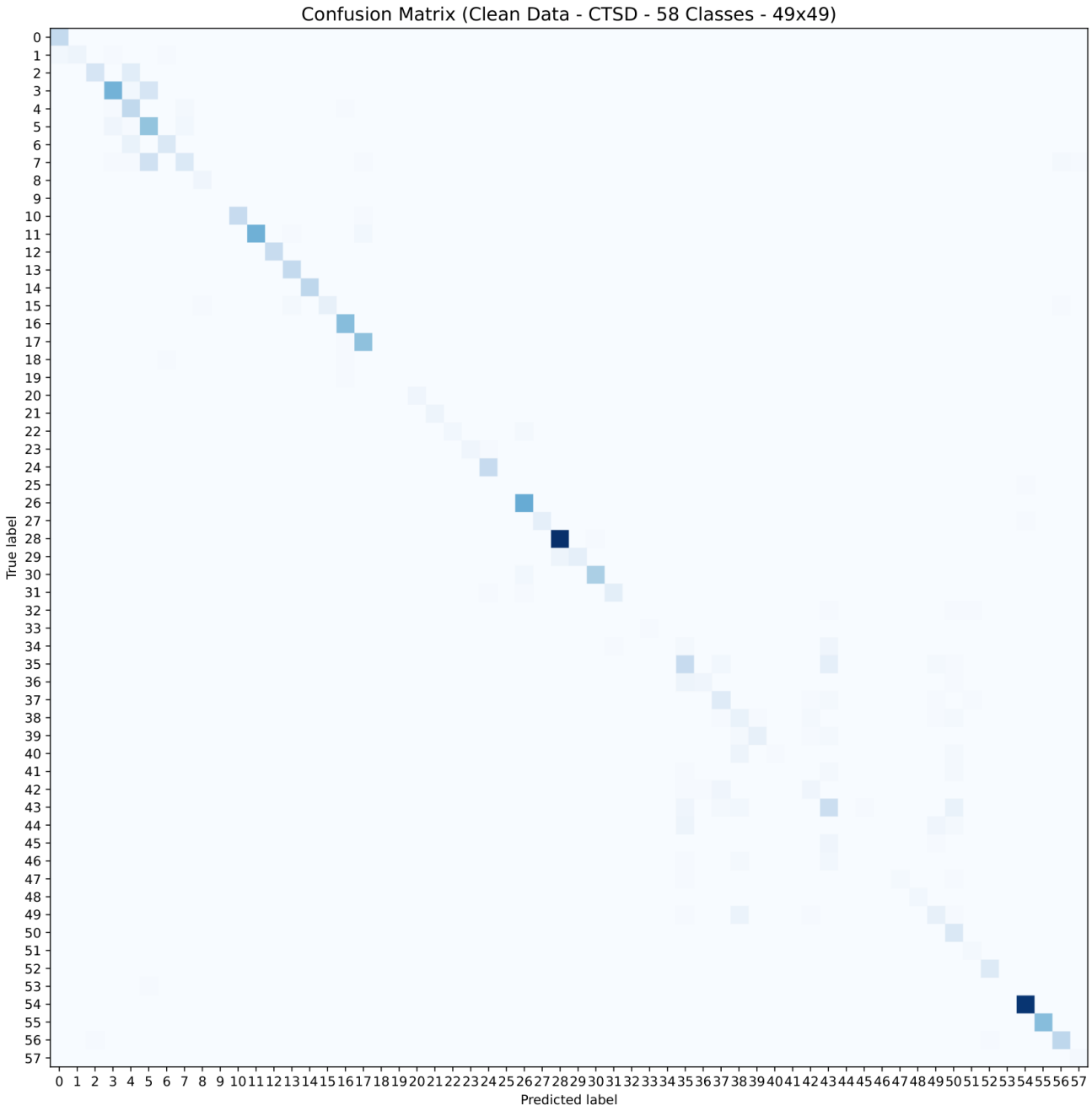
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/baseline/btsd/test_without_noise_btsd.pdf | SHA-256 prefix: 92149395b674

Additional corruption matrices are saved separately in output/baseline/btsd/noise/ (14 severity-5 PDFs).

Baseline CNN | CTSD confusion matrix

CLEAN TEST DATA | 58 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

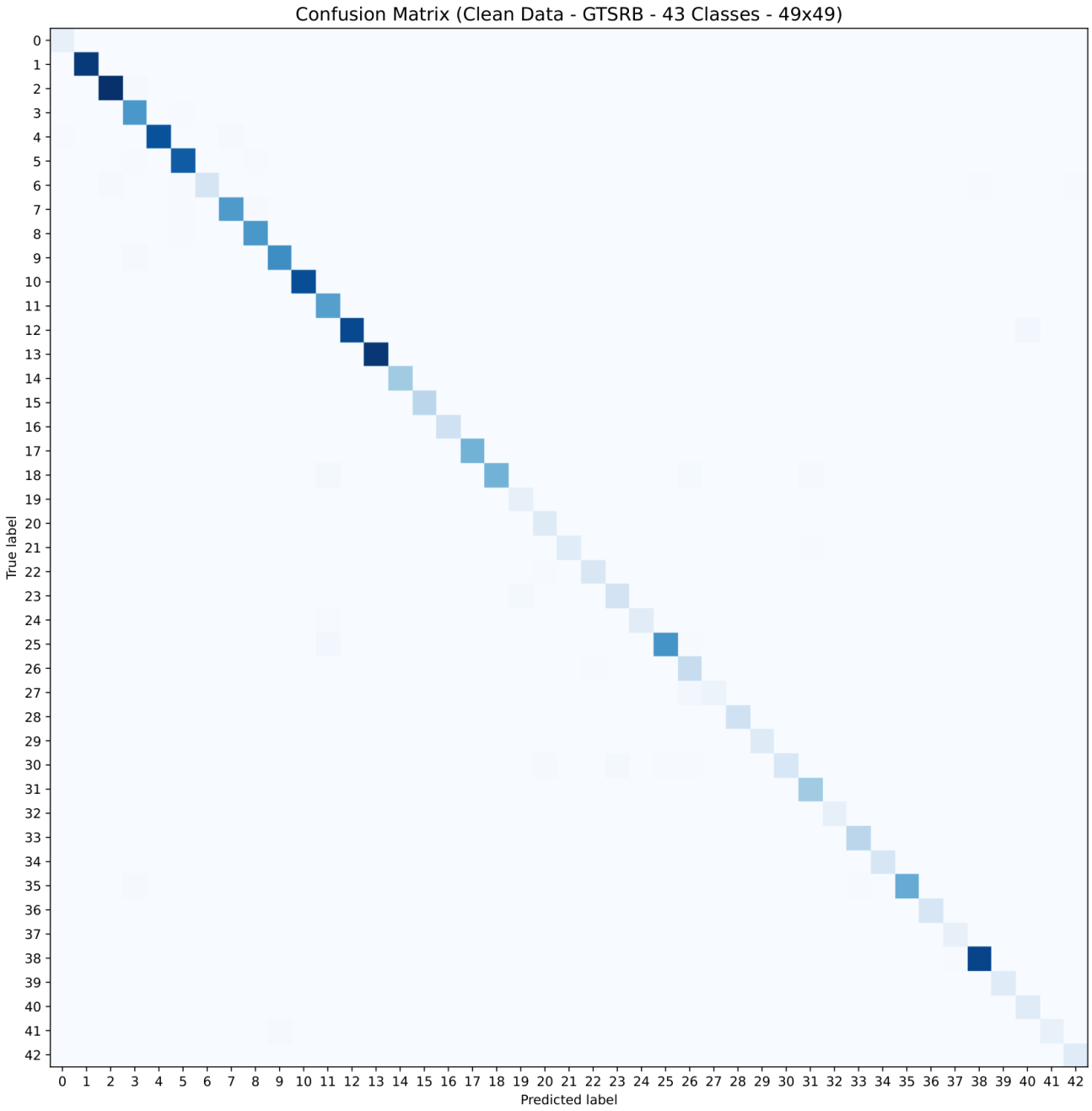
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/baseline/ctsd/test_without_noise_ctsd.pdf | SHA-256 prefix: 058f03ac895f

Additional corruption matrices are saved separately in output/baseline/ctsd/noise/ (14 severity-5 PDFs).

CNN + AdaBN | GTSRB confusion matrix

CLEAN TEST DATA | 43 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

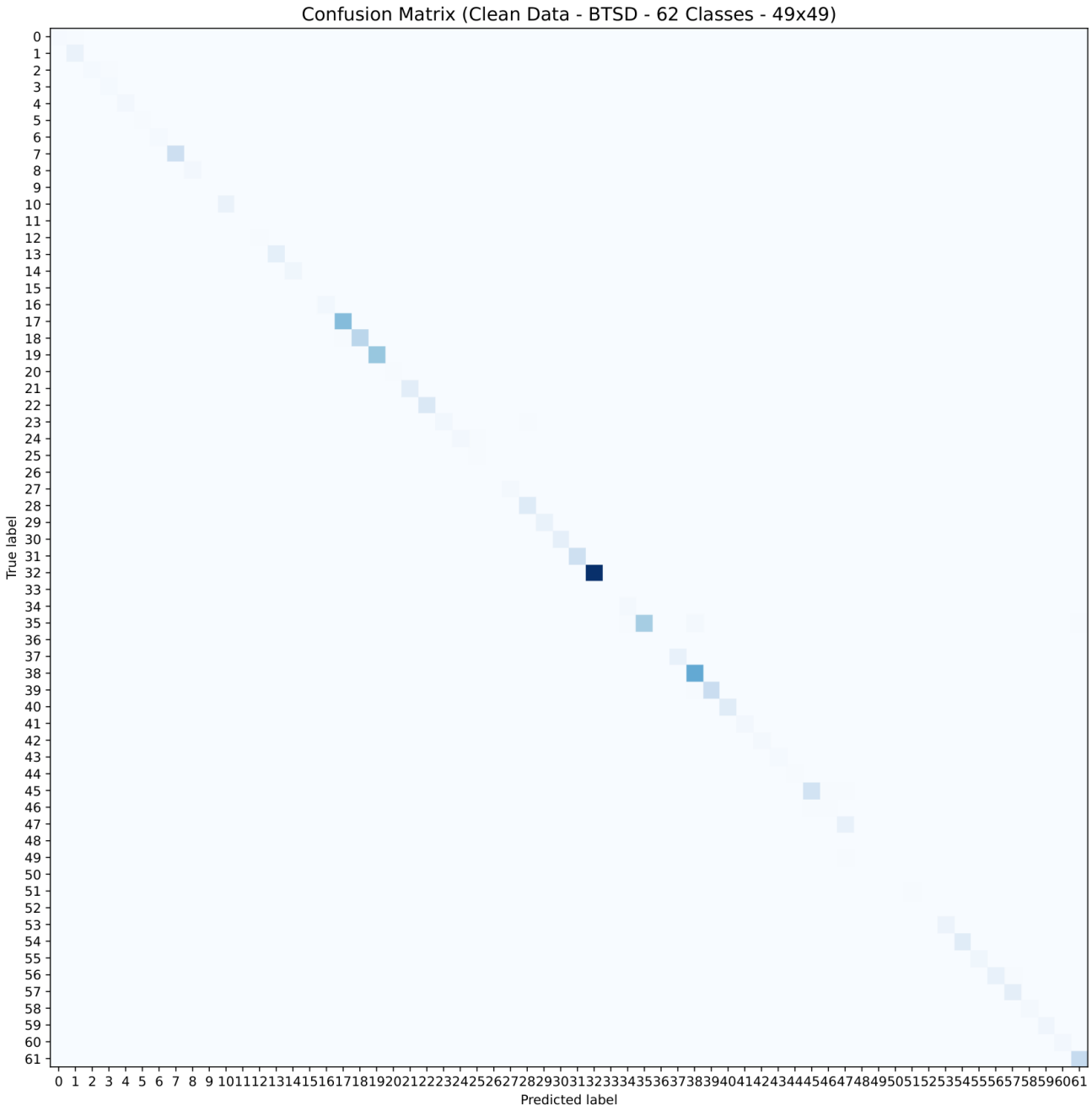
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/adabn/gtsrb/test_without_noise_gtsrb.pdf | SHA-256 prefix: d7631ad7e9f1

Additional corruption matrices are saved separately in output/adabn/gtsrb/noise/ (14 severity-5 PDFs).

CNN + AdaBN | BTSD confusion matrix

CLEAN TEST DATA | 62 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

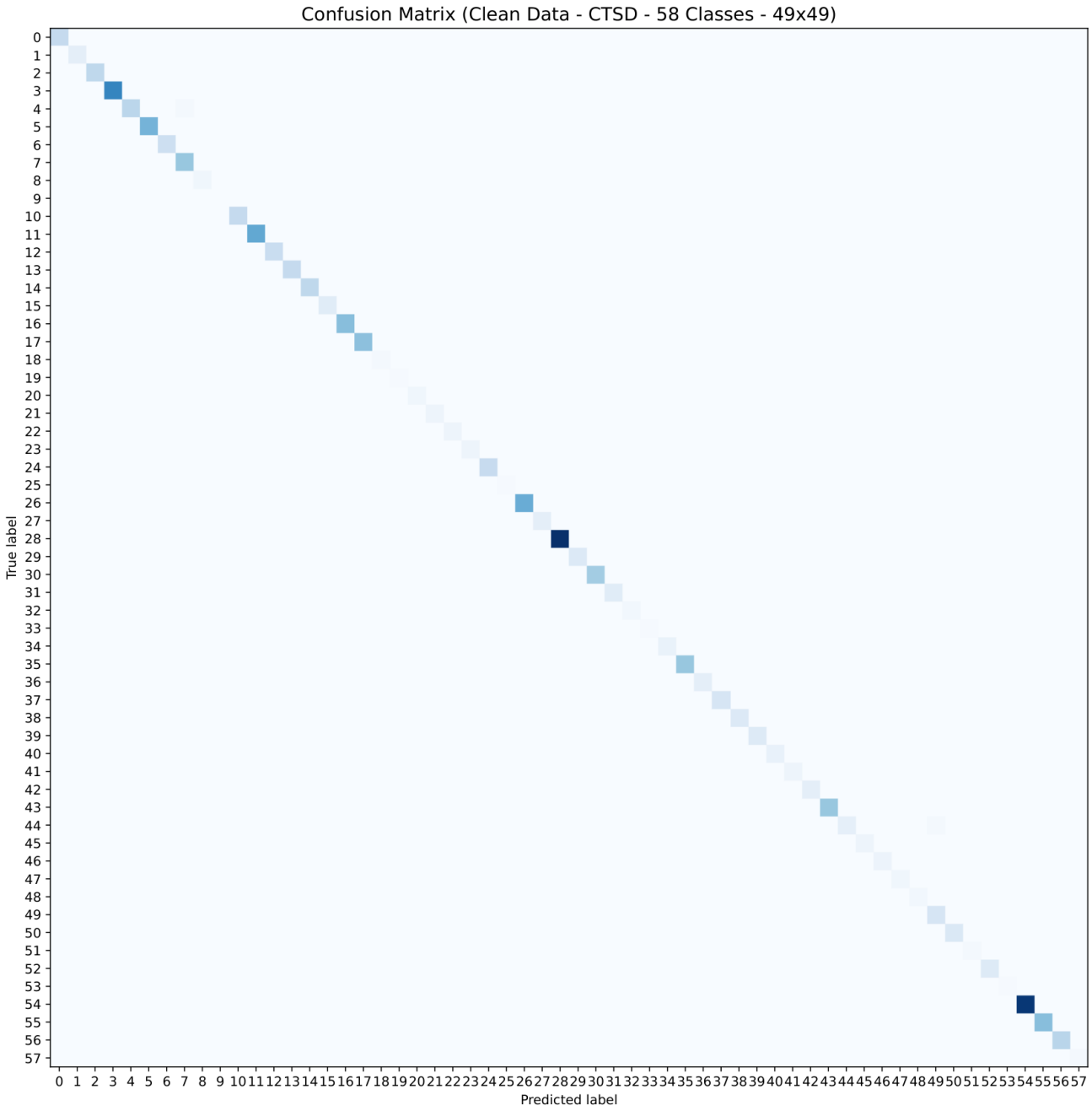
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/adabn/btsd/test_without_noise_btsd.pdf | SHA-256 prefix: 12e43c20a1d7

Additional corruption matrices are saved separately in output/adabn/btsd/noise/ (14 severity-5 PDFs).

CNN + AdaBN | CTSD confusion matrix

CLEAN TEST DATA | 58 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

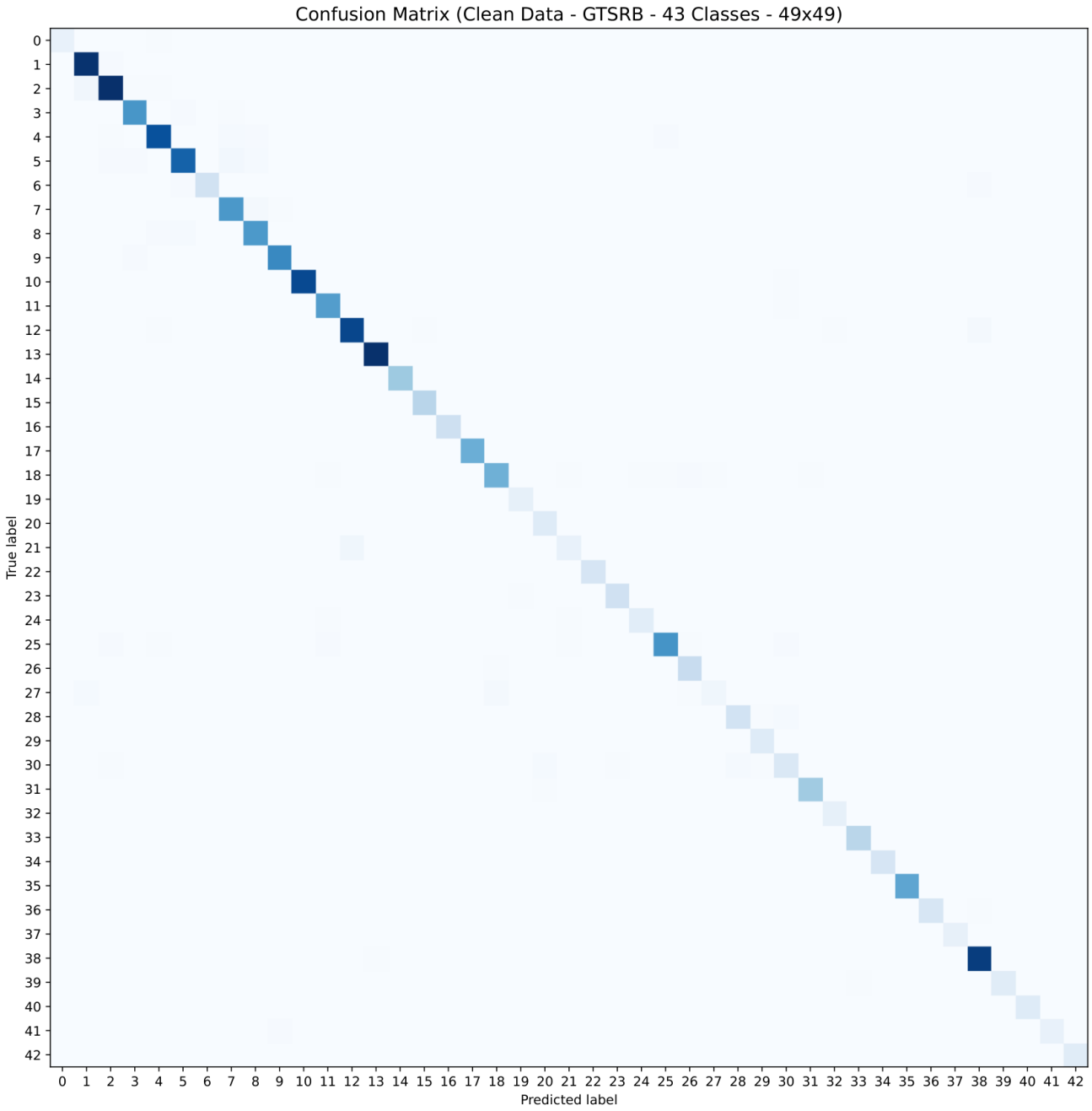
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/adabn/ctsd/test_without_noise_ctsd.pdf | SHA-256 prefix: f23235fb82bc

Additional corruption matrices are saved separately in output/adabn/ctsd/noise/ (14 severity-5 PDFs).

CNN + ACO | GTSRB confusion matrix

CLEAN TEST DATA | 43 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

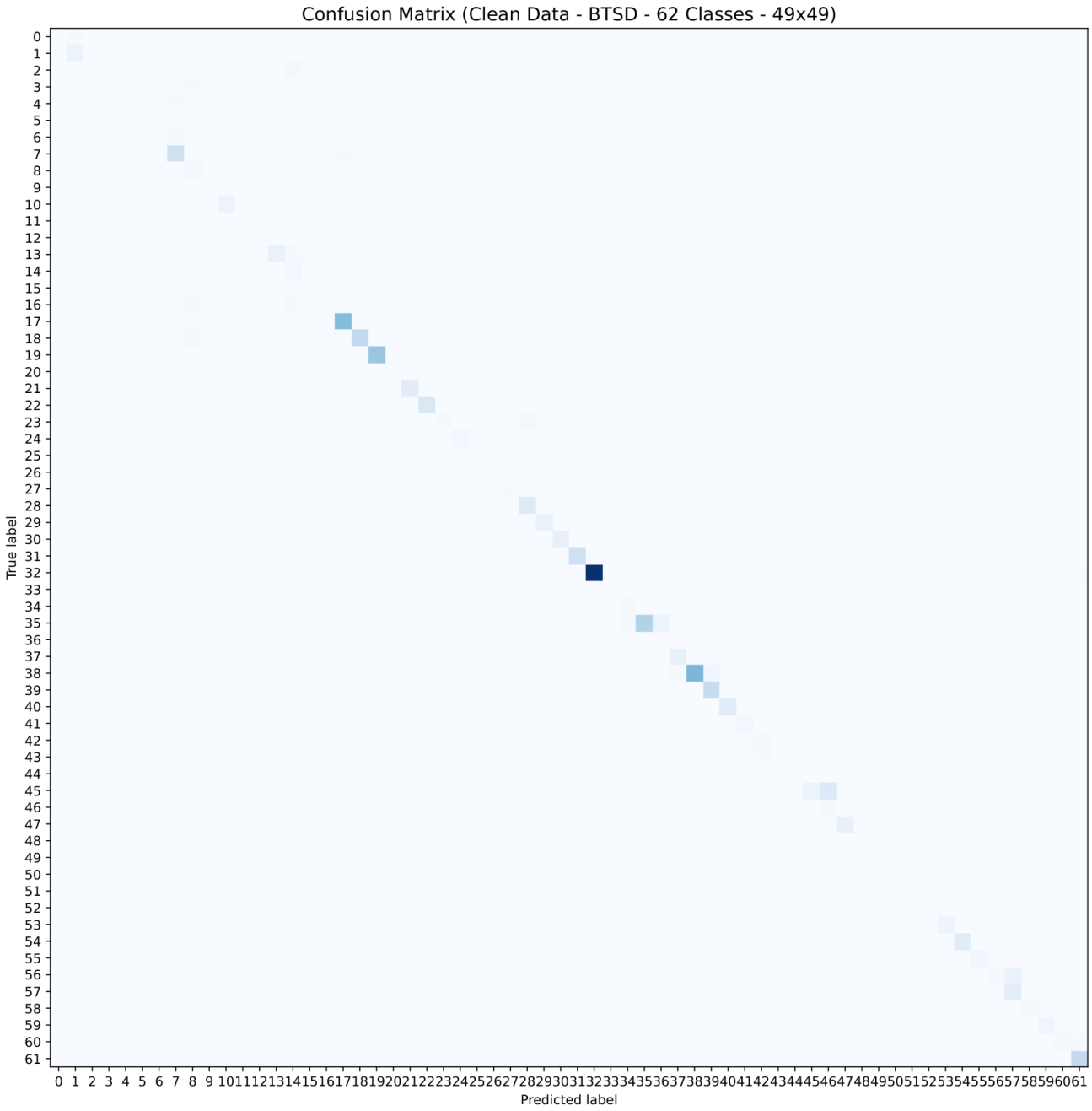
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/aco/gtsrb/test_without_noise_gtsrb.pdf | SHA-256 prefix: cb32d783b92f

Additional corruption matrices are saved separately in output/aco/gtsrb/noise/ (14 severity-5 PDFs).

CNN + ACO | BTSD confusion matrix

CLEAN TEST DATA | 62 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

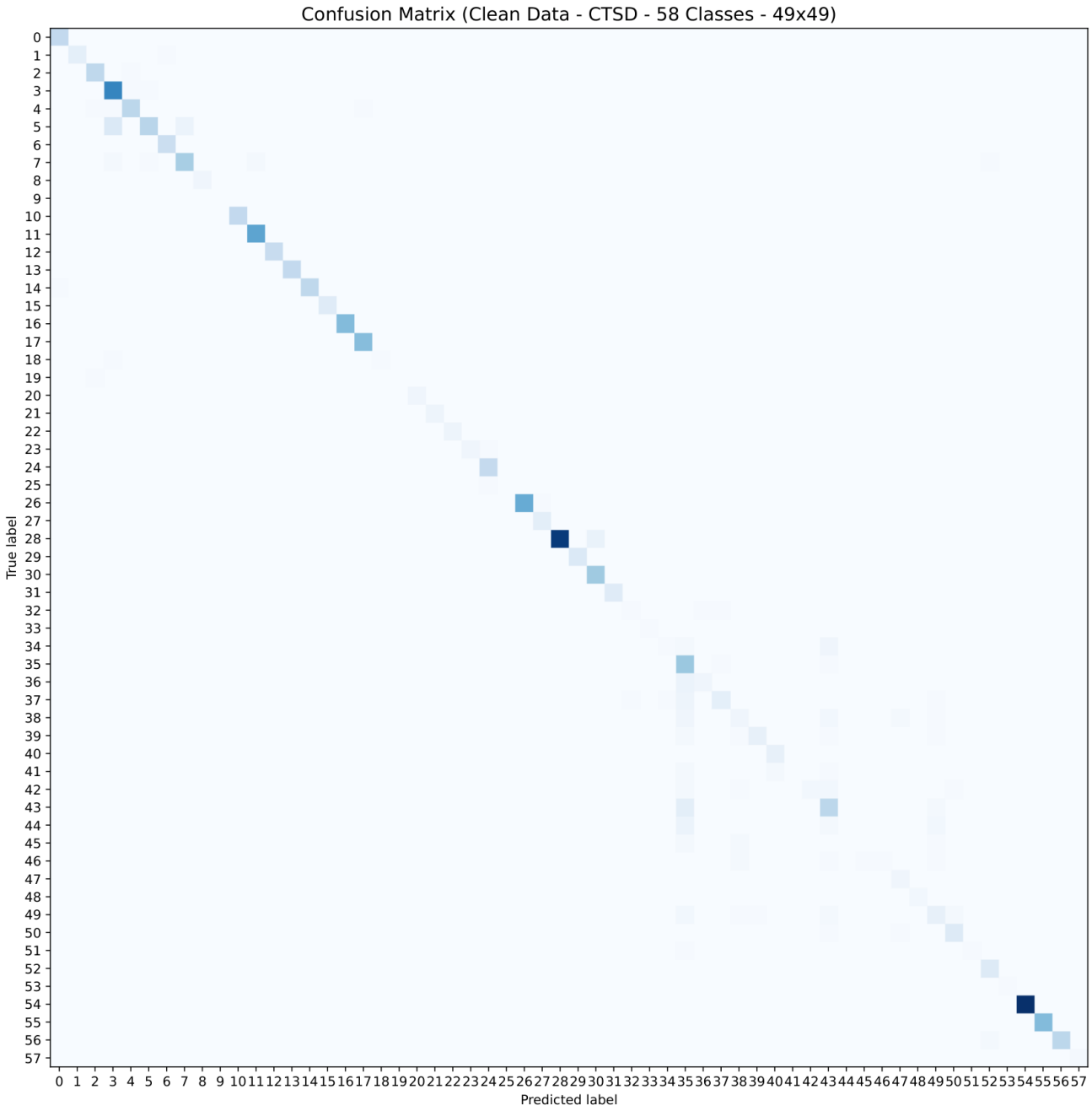
This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/aco/btsd/test_without_noise_btsd.pdf | SHA-256 prefix: d63d4896a20d

Additional corruption matrices are saved separately in output/aco/btsd/noise/ (14 severity-5 PDFs).

CNN + ACO | CTSD confusion matrix

CLEAN TEST DATA | 58 CLASSES | 49 x 49 RGB INPUT



True class: vertical axis. Predicted class: horizontal axis. Diagonal = correct; off-diagonal = misclassified. Colors encode raw counts.

This figure is from clean testing. It does not represent the 70-condition corruption average shown in the numerical tables.

Source: output/aco/ctsd/test_without_noise_ctsd.pdf | SHA-256 prefix: a1763ce392bf

Additional corruption matrices are saved separately in output/aco/ctsd/noise/ (14 severity-5 PDFs).